HOMEWORK 3

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6.1. Prove, that if $A : V \to W$ is an isomorphism and $v_1, v_2, \ldots, v_n$ is a basis in V, then $Av_1, Av_2, \ldots, Av_n$ is a basis in W.

Proof. Let $v \in V$ and $w \in W$ be arbitrary vectors. Then, $v = \sum_{i=1}^{n} a_i v_i$. Then, applying the transformation, $w = Av = A\sum_{i=1}^{n} a_i v_i$. Thus, we see $w = Av$ spans W since w was arbitrary. V's basis is linearly independent by definition. Thus, $\sum_{i=1}^{n} a_i v_i = 0 \implies \sum_{i=1}^{n} |a_i| = 0$. Using this, $A\sum_{i=1}^{n} a_i v_i = 0w$. We know $\sum_{i=1}^{n} a_i v_i$ is linearly independent, therefore $A\sum_{i=1}^{n} a_i v_i$ must be a basis for W. □

6.2. Find all right inverses to the 1×2 matrix $A = [1, 1]$. Conclude that A is not left invertible.

Proof. Let B be an arbitrary right inverse. $\begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}b_1 \\ b_2\end{bmatrix} = \begin{bmatrix}b_1 + b_2\end{bmatrix}$. From this, we must show that $b_1 + b_2 = 1$. We show $b_1 = 1 - b_2$. Thus, B is of the form of $\begin{bmatrix}1 - b_2 \\ b_2\end{bmatrix}$. A cannot have a left inverse, because it would produce a 2×2 matrix of the form $\begin{bmatrix}b_1 & b_1 \\ b_2 & b_2\end{bmatrix}$. Since $b_1, b_2 \in \mathbb{R}$, there exists an $n \in \mathbb{R}$ such that $b_1 = nb_2$. Therefore, the matrix is not linearly independent, showing it cannot be inverted. □

6.3. Find all left inverses of the column $(1, 2, 3)^T$

Proof. Yes, $\begin{bmatrix}1 - 2b_2 - 3b_3 & b_2 & b_3\end{bmatrix}$. We derive this from $BA = I$. □

6.4. Is the column $(1, 2, 3)^T$ right invertible? Justify.

Proof. No. If you perform matrix multiplication, you will get a 3×3 matrix that is linearly dependent, which wouldn't work since I is linearly independent. □

6.5. Find two matrices A and B that AB is invertible, but A and B are not.

Proof. $A = \begin{bmatrix}1 & 0\end{bmatrix}$, $B = \begin{bmatrix}1 \\ 0\end{bmatrix}$, $AB = \begin{bmatrix}1\end{bmatrix}$ □

6.6. Suppose the product AB is invertible. Show that A is right invertible and B is left invertible.

Proof. Let $AB : V \to W$. Since AB is invertible, define $(AB)^{-1} : W \to V$. Since AB is invertible, that means $(AB)^{-1}AB = AB(AB)^{-1} = I$. Using associativity rules, we see that A has a right inverse and B has a left inverse. □

6.7. Let A and AB be invertible. Prove that B is invertible.

Proof. Let $(AB)^{-1}$ be the inverse of AB and A^{-1} be the inverse for A. We show B is left invertible by associativity: $(AB)^{-1}AB = ((AB)^{-1}A)B = I$. To show B is right invertible, $A(B(AB)^{-1}) = I \to B(AB)^{-1} = A^{-1}$. Since A is invertible, $B((AB)^{-1}A) = I$. Thus, B is invertible. □

6.8. Let A be $n \times n$ matrix. Prove that if $A^2 = 0$ then A is not invertible.

Proof. Assume for contradiction that A is invertible. We may rewrite $A^2 = AA = 0$. Applying invertibility, $A = 0A^{-1} = 0$. $A = 0$ is a contradictino, since the zero vector is not invertible. Therefore, A is not invertible. □

6.9. Suppose $AB = 0$ for some non-zero matrix B. Can A be invertible? Justify.

Proof. No. Assume A is invertible. $AB = 0 \to B = 0A^{-1} = 0$. $B \neq 0$, contradiction. Therefore A isn't invertible. □

6.10. Write matrices of the linear transformations T_1 and T_2 in $\mathbb{F}^5$, defined as follows: T_1 interchanges the coordinates x_2 and x_4 of the vector x, and T_2 just adds to the coordinate x_2 a times the coordinate x_4, and does not change other coordinates.

$$T_1 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_4 \\ x_3 \\ x_2 \\ x_5 \end{pmatrix}, \qquad T_2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 + \alpha x_4 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix};$$

Show that T_1 and T_2 are invertible transformations, and write the matrices of the inverses.

Proof. By looking, we see that $T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ and $T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \alpha & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$.
Firstly, notice these are 5×5 matrices, so they meet that condition. T_1 is invertible because we can perform a row swap between R_2 and R_4, giving us I. For T_2, we can multiply R_4 by α, and let $R_2 = R_2 - \alpha R_4$, thus giving us the identity matrix. Thus, both of these are invertible. By augmenting our T_1 and T_2 with I, we see that $T_1 = T_1^{-1}$. and $T_2^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\alpha & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ □